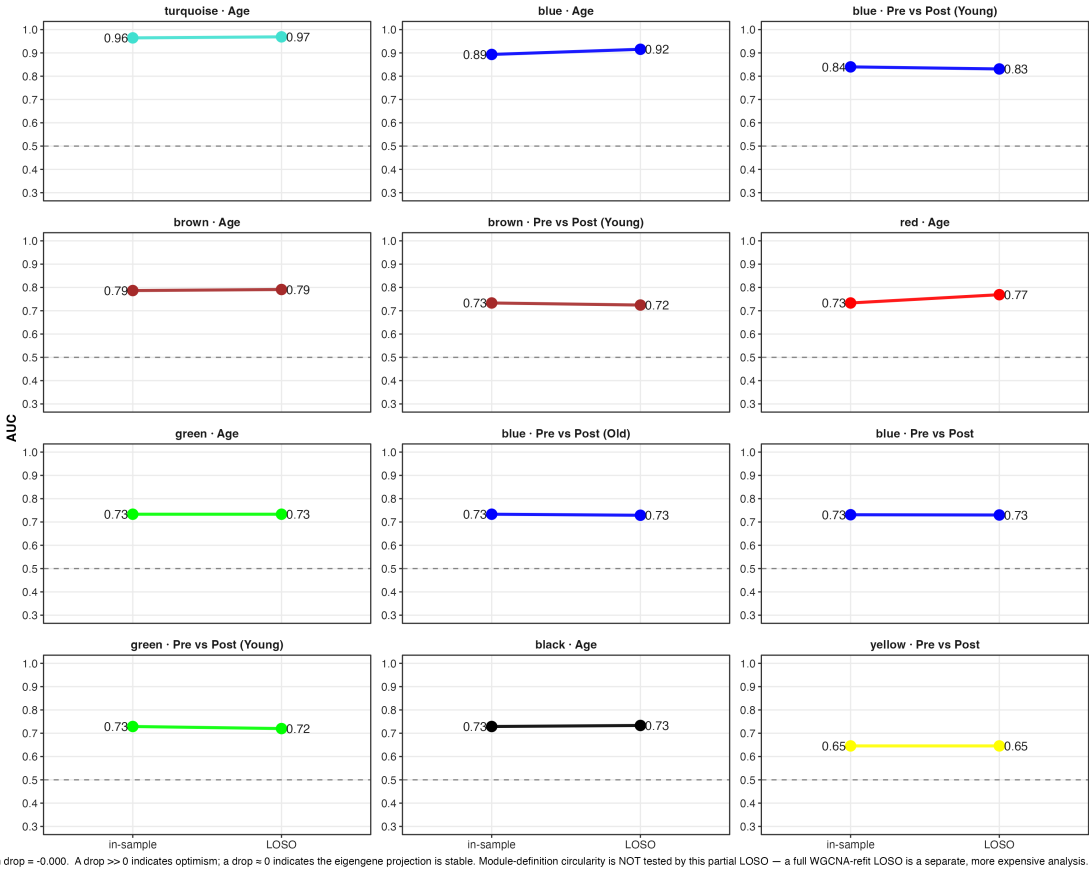


**A** F07 Supp — LOSO sensitivity on top-12 module-outcome AUCs

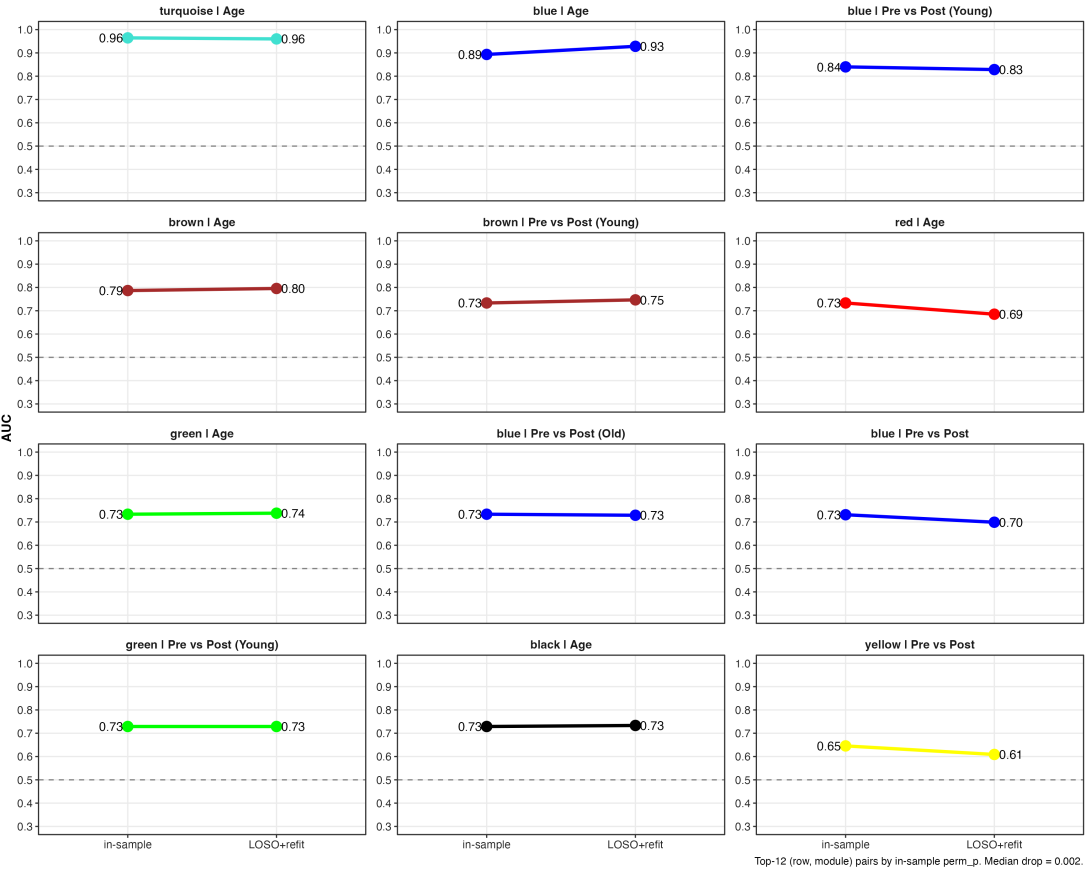
Module assignments held fixed (full-sample WGCNA); eigengene PC refit per LOSO fold and held-out subject projected onto training PC.



Median drop = -0.000. A drop >> 0 indicates optimism; a drop ≈ 0 indicates the eigengene projection is stable. Module-definition circularity is NOT tested by this partial LOSO — a full WGCNA-refit LOSO is a separate, more expensive analysis.

**B** F07 Supp — Full WGCNA-refit LOSO sensitivity

Network refit on n-1 subjects per fold (power=12, same parameters as full-sample). Training modules matched to full-sample modules by Jaccard overlap of protein assignments; held-out subject projected onto training PC.



Top-12 (row, module) pairs by in-sample perm\_p. Median drop = 0.002.

**C**

